Results and Comparison

AI Research Copilot | Project report section

1. System results

The project uses the pretrained **sentence-transformers/all-MiniLM-L6-v2** model to convert research-paper passages and user questions into 384-dimensional embeddings. Vectors are normalized and searched with a FAISS inner-product index. The local seed manifest specifies seven sample papers and 173 expected indexed text chunks. These are reproducible indexing results, not an accuracy score.

Item	Observed project result
Embedding model	all-MiniLM-L6-v2; pretrained; 384-dimensional sentence embeddings
Seed corpus	7 sample research papers
Indexed passage vectors	173 expected FAISS vectors, according to seed_manifest.json
Retrieval method	Normalized embeddings with FAISS inner-product search; retrieved passages ground research answers
Generation model	qwen2.5:3b via Ollama; a separate text-generation component, not an embedding baseline

2. Accuracy and evaluation status

Project-specific accuracy: Not measured. This repository uses a pretrained embedding model; it does not train a project-specific classifier or include a labelled test set of research questions and relevant passages. A percentage such as "95% accuracy" would not be supported by the available evidence. The 173 indexed chunks measure corpus/index size, not correctness.

For a defensible retrieval evaluation, prepare held-out research questions and manually mark relevant passages. Run each model against the same corpus and settings, then report Recall@5/10, Mean Reciprocal Rank (MRR), and nDCG@10. For paper-category classification, use a labelled test set and report accuracy, precision, recall, and F1-score.

3. Model comparison

The comparison uses published model characteristics, not measurements from this project. The official Sentence Transformers guide describes all-mpnet-base-v2 as the higher-quality general-purpose option and all-MiniLM-L6-v2 as approximately five times faster. Accuracy on this project cannot be compared until both models are tested on the same labelled set.

Model	Vector size	Published comparison	Project role
all-MiniLM-L6-v2 **(current)**	384	Compact and fast; described as about 5x faster than all-mpnet-base-v2.	Current paper/query embedding model.
all-mpnet-base-v2	768	Higher-quality general-purpose alternative per the official guide; larger embedding size.	Candidate for a controlled local retrieval test.

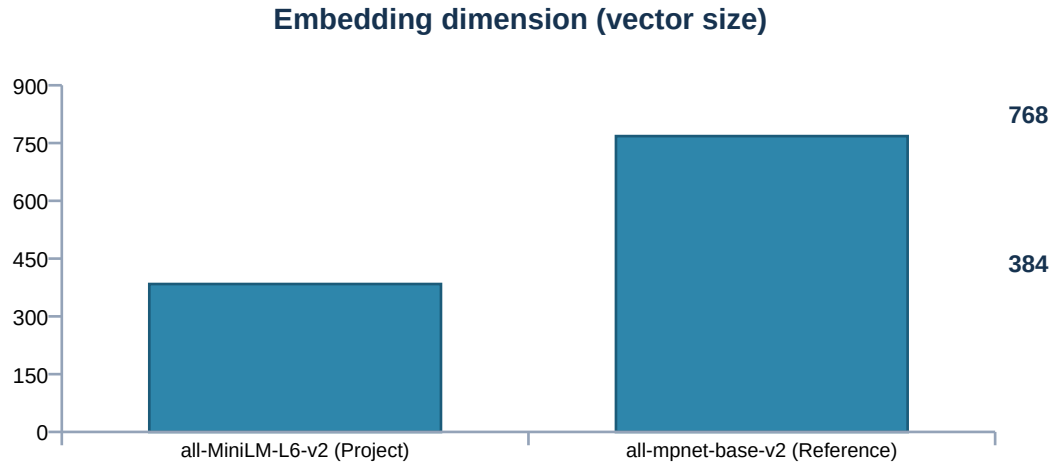

Figure 1. Published embedding dimensions of the current model and a general-purpose reference model. This chart compares representation size, not accuracy. Dimensions are taken from the model cards.

4. Discussion and conclusion

The current MiniLM choice offers a practical balance for a local research assistant: 384-dimensional vectors reduce storage and search cost while supporting semantic retrieval beyond exact keyword matches. all-mpnet-base-v2 uses 768-dimensional vectors and may provide stronger general-purpose semantic quality at higher compute and storage cost. These published characteristics do not establish which model is more accurate on this project's papers.

The verified project result is indexing the seven-paper seed set into 173 expected vectors and retrieving evidence from the indexed passages. No project-specific accuracy, precision, recall, or F1 result is available yet. A fair final comparison requires the same labelled query set, corpus, and retrieval settings for both models.

5. References

[1] Sentence Transformers, "Pretrained Models," official documentation. General-purpose model comparison and relative speed statement. https://www.sbert.net/docs/sentence_transformer/pretrained_models.html

[2] Hugging Face model card, sentence-transformers/all-MiniLM-L6-v2. 384-dimensional embeddings and semantic-search use. https://huggingface.co/sentence-transformers/all-MiniLM-L6-v2

[3] Hugging Face model card, sentence-transformers/all-mpnet-base-v2. 768-dimensional embeddings and semantic-search use. https://huggingface.co/sentence-transformers/all-mpnet-base-v2

[4] Project seed_manifest.json and backend/config.py. Local configuration: seven seeded paper records, 173 expected vectors, and the all-MiniLM-L6-v2 embedding model.